

Hyperparameter Tuning for Hybrid SVM-KD Tree Qubit Readout

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Theoretical Framework for Hybrid Hyperparameter Automation

This document details the mathematical framework for automating hyperparameter selection within a hybrid Active Learning (AL) architecture. The system optimizes the interplay between a Global Model (Support Vector Machine) for high-throughput classification and a Local Model (KD-Tree) for noise-resilient refinement in the I-Q plane.

The S-Ratio: A Metric for Cluster Overlap

The fundamental challenge in superconducting qubit readout is the discrimination of multi-qubit states, which manifest as Gaussian clusters in the complex I-Q (In-phase and Quadrature) plane. To quantify the difficulty of this classification task, we define the S-Ratio, a metric inspired by the Fisher Criterion used in Linear Discriminant Analysis (LDA).

Relationship to the Fisher Criterion

The Fisher Criterion, $J(w)$, quantifies the separability of two classes by comparing the distance between their means to their internal scatter:

$$J(w) = \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}$$

A higher $J(w)$ indicates superior discriminability. Conversely, our proposed S-Ratio measures the inverse relationship—focusing on the degree of overlap or ambiguity between states.

For a specific qubit state cluster i with covariance matrix Σ_i and its nearest neighboring cluster j with centroid μ_j , the S-ratio is defined as:

$$S_i = \frac{Tr(\Sigma_i)}{\min_{j \neq i} \|\mu_i - \mu_j\|^2}$$

Where:

$Tr(\Sigma_i)$: The trace of the covariance matrix, representing the total variance ($\sigma_I^2 + \sigma_Q^2$) of the state.

$\|\mu_i - \mu_j\|^2$: The squared Euclidean distance between the centroids of cluster i and its nearest neighbor j .

We can compute the aggregate system difficulty, $\bar{S}$, is the mean of these ratios across all N clusters:

$$\bar{S} = \frac{1}{N} \sum_{i=1}^N S_i$$

$S \ll 1$ (High Separability): Clusters are "point-like" and far apart. The SVM can easily find a clear decision boundary.

$S \approx 1$ or higher (High Ambiguity): The "blurriness" (variance) is comparable to or greater than the distance between states. Clusters merge into a single blob, requiring local geometric intervention.

Threshold Selection

The SVM confidence threshold should be inversely proportional to S . If clusters are overlapping (S is high), we need a wider "ambiguity zone" to let the KD-tree assist.

K-Neighbors Selection

As variance increases relative to distance, the local density becomes noisier. A higher k is required to "smooth" the local geometric label.

Low S (High separation): The model suggests a lower threshold. On hardware, this translates to bypassing the KD-tree logic most of the time, saving power and clock cycles.

High S (Noisy overlap): The model suggests a high threshold and higher k . The algorithm should trigger the KD-tree search more frequently, using the "Local Geometry" to resolve ambiguity that the global SVM hyperplane cannot handle.

This hyperparameter selection is implemented in Python with the following (detailed) approach.

SVM Ambiguity Threshold The threshold T defines the confidence level below which the SVM delegates a sample to the KD-Tree. As overlap increases, we widen the "ambiguity zone":

$$T = \min(0.95, 0.7 + 0.2 \cdot \bar{S})$$

KD-Tree Neighbor Selection(k) To ensure a stable local majority vote in high-noise environments, the number of neighbors k scales with the S -ratio:

$$k = \lfloor 5 + 10 \cdot \bar{S} \rfloor$$

(Note: k is adjusted in the implementation to the nearest odd integer to prevent ties in classification).

Min-Advantage Criterion for Trust Calibration The "Min Advantage" parameter (α) acts as a gating mechanism. If the I-Q distribution is clean (low $\bar{S}$), the KD-Tree must demonstrate a higher performance gain over the SVM to be utilized:

$$\alpha = \max(0.05, 0.2 - 0.1 \cdot \bar{S})$$

We only update the model if the KD-Tree demonstrates a significant performance lift over the SVM on the ambiguous batch:

$$\Delta_{acc} = Acc(Y_{KD}) - Acc(Y_{SVM})$$

The update triggers only if $\Delta_{acc} \geq min_advantage$.

Results: <https://docs.google.com/document/d/12ajxhXYuXWn4haq0uT4EqqI8ykt0-SVrmjalguH-Ya8/edit?usp=sharing>